

Lecture on Logical Agents: Knowledge-Based Agents

Course: Artificial Intelligence

Instructor: Syed Hamed Raza

Overview of the Lecture:

This lecture introduces the concept of **Logical Agents** and **Knowledge-Based Agents** in artificial intelligence. These agents use logical reasoning to make decisions based on their knowledge about the world. We will explore how logical agents represent knowledge, how they reason about it, and how they make decisions by drawing inferences from the knowledge base. By the end of the lecture, you will have an understanding of how logical agents operate and their applications in AI systems.

Agenda:

1. **Introduction to Logical Agents**
 2. **Knowledge-Based Agents**
 3. **Representation of Knowledge**
 - **Propositional Logic**
 - **First-Order Logic**
 4. **Reasoning with Logic**
 - **Inference**
 - **Forward and Backward Chaining**
 5. **Applications of Knowledge-Based Agents**
 6. **Classroom Discussion and Problem Solving**
-

1. Introduction to Logical Agents

What are Logical Agents?

- **Logical agents** are intelligent systems that use logic to represent their knowledge about the world and to make decisions based on that knowledge.
- These agents are capable of **logical reasoning**, which involves applying inference rules to their knowledge base to derive new facts and make decisions.

Key Characteristics of Logical Agents:

- **Knowledge Representation:** Logical agents use formal languages like **propositional logic** or **first-order logic** to represent knowledge.
- **Inference Mechanism:** Logical agents use **inference rules** to derive new information from their existing knowledge base. Inference helps the agent reason about the world and take actions accordingly.

- **Goal-Oriented:** Logical agents act based on goals they wish to achieve, deriving actions from their knowledge and reasoning about how to achieve those goals.
-

2. Knowledge-Based Agents

What are Knowledge-Based Agents?

- A **knowledge-based agent** is an agent that makes decisions based on a knowledge base (KB), which contains a set of facts and rules that describe the environment and the agent's beliefs about it.
- The knowledge base is often augmented with **inference mechanisms** that allow the agent to derive new information from what it already knows.

Components of a Knowledge-Based Agent:

- **Knowledge Base (KB):** A structured set of knowledge that describes the agent's environment, including facts and relationships. The knowledge can be represented in the form of logical statements.
 - **Inference Engine:** A component that uses the knowledge base to derive new facts or make decisions based on reasoning. It applies rules of inference to extract conclusions from the knowledge base.
 - **Decision-Making Mechanism:** Based on the conclusions drawn by the inference engine, the agent selects actions that lead to goal achievement.
-

3. Representation of Knowledge

Propositional Logic (PL):

- **Propositional logic** is a simple formal language used to represent knowledge. It involves statements that can be either **true** or **false**.
- In propositional logic, we use **propositions** (such as "It is raining" or "The light is on") and **logical connectives** (AND, OR, NOT, IF-THEN) to create logical expressions.

Example:

- Let **P** represent "It is raining" and **Q** represent "The ground is wet."
- The statement "If it is raining, then the ground is wet" is written as:
 $P \rightarrow Q$ (If P, then Q).

First-Order Logic (FOL):

- **First-order logic** (also known as **predicate logic**) is a more expressive formalism than propositional logic. It allows for the use of **quantifiers** (e.g., "all," "some") and **predicates** (properties or relationships between objects) to express knowledge.

Example:

- "All humans are mortal" is written in first-order logic as:
 $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ (For all x, if x is a human, then x is mortal).
- **Objects:** Individuals in the world (e.g., John, Mary, the table).

- **Predicates:** Functions that return true or false based on objects (e.g., $\text{Human}(x)$, $\text{Mortal}(x)$).
 - **Quantifiers:** Describes how many objects satisfy a condition (e.g., $\forall$ for "all," $\exists$ for "there exists").
-

4. Reasoning with Logic

Inference in Logic:

- **Inference** is the process of deriving new knowledge from existing knowledge using **logical rules**. The inference engine applies these rules to the knowledge base to derive new facts.
- **Example of Inference:**
 - If we know that "All humans are mortal" ($\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$) and "Socrates is a human" ($\text{Human}(\text{Socrates})$), we can infer that "Socrates is mortal" ($\text{Mortal}(\text{Socrates})$) using **Modus Ponens** (If $P \rightarrow Q$ and P , then Q).

Forward and Backward Chaining:

- **Forward Chaining:** Starts with the known facts and applies inference rules to generate new facts until a goal is reached. It is often used in **data-driven reasoning**.
 - **Example:** If you know that "A dog is an animal" and "All animals need food," forward chaining would deduce that "A dog needs food."
 - **Backward Chaining:** Starts with a goal and works backward to find the facts that support that goal. It is typically used in **goal-driven reasoning**.
 - **Example:** If the goal is to prove that "Socrates is mortal," backward chaining would check if "Socrates is human" (which is known) and then use the rule "All humans are mortal" to conclude that Socrates is mortal.
-

5. Applications of Knowledge-Based Agents

Expert Systems:

- **Expert systems** are a type of knowledge-based agent designed to mimic the decision-making abilities of a human expert in a specific domain. They use a knowledge base of facts and rules to make decisions or solve problems.
- **Example:** A medical expert system may be used to diagnose diseases based on symptoms. It uses a knowledge base of medical facts and rules to infer possible diagnoses.

Robotics:

- **Robots** often use knowledge-based systems to make decisions based on their environment. They use sensors to collect data and infer knowledge about their surroundings to plan actions.

Natural Language Processing (NLP):

- **NLP systems** can also be knowledge-based, where the agent reasons about the meaning of sentences or text based on a knowledge base of linguistic rules and facts.
-

6. Classroom Discussion and Problem Solving

- **Open Discussion:** How do knowledge-based agents differ from reactive agents? Can you think of any real-world applications of knowledge-based agents that you interact with daily?
 - **Hands-on Activity:** Implement a simple knowledge-based agent in Python that uses **forward chaining** to derive conclusions based on a set of rules. For example, a system that infers whether a person is eligible for a loan based on certain criteria.
-

Conclusion:

- **Summary:** In this lecture, we explored the concept of **Logical Agents** and **Knowledge-Based Agents**. We discussed how logical agents use logical reasoning and inference to make decisions based on their knowledge base. We also covered how knowledge is represented in **propositional logic** and **first-order logic** and the mechanisms of **forward and backward chaining** for reasoning.
- **Next Steps:** In the next lecture, we will explore **automated reasoning** and delve deeper into more advanced topics in logic-based AI, such as **theorem proving** and **knowledge representation in uncertain environments**.